

Implementing Secure and Controlled Web Access Using Tor, Privoxy, and Windows Firewall

Prepared by: Melanie Seguin
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Overview

This lab provides hands-on experience in building a secure, privacy-focused browsing environment using three core tools: the Tor anonymization network, the Privoxy proxy server, and Windows Defender Firewall. The goal was to simulate a controlled forensic lab environment where all web traffic is routed through the Tor network, only specific applications are permitted internet access, and all traffic is logged and monitored in real time. The lab reinforces practical knowledge in network anonymization, proxy configuration, firewall policy enforcement, and traffic verification.

Situation

As a cybersecurity technician working in a forensic lab, internet usage must be both anonymized and strictly controlled. The organization requires that all outbound web traffic passes through the Tor network, and that only authorized applications such as the Tor Browser and Privoxy proxy are permitted to reach the internet. Standard browsers operating outside the proxy chain must be blocked at the firewall level. The environment was built using Windows 11 running inside a Parallels Desktop virtual machine on an Apple Silicon MacBook.

System Architecture

The solution was built around a three-layer architecture in which each component handles a specific role in the traffic chain. The Tor Browser establishes an anonymous connection to the Tor network and listens locally on port 9150. Privoxy acts as an HTTP proxy running on port 8118, filtering incoming browser requests and forwarding them to Tor. Firefox is configured to route all traffic through Privoxy. Windows Defender Firewall enforces application-level access control, ensuring that Firefox cannot bypass the proxy chain and reach the internet directly.

The complete traffic flow runs as follows: Firefox sends a request to Privoxy on port 8118. Privoxy receives the request, applies its filtering rules, and forwards it to Tor on port 9150. Tor encrypts the

traffic and routes it through a series of volunteer-operated relay nodes before it reaches the destination server. The destination sees only the IP address of a Tor exit node, not the real IP address of the lab machine.

Step 1: Tor Browser Installation and Configuration

The Tor Browser version 15.0.11 was downloaded from the official Tor Project website and installed on the Windows 11 virtual machine. Upon launch, the browser connected successfully to the Tor network, displaying the confirmation screen indicating it was ready for private browsing. The Tor Browser operates an internal SOCKS proxy on localhost port 9150, which serves as the entry point for all anonymized traffic forwarded from Privoxy. Tor achieves anonymization by routing traffic through at least three relay nodes before reaching the destination, with each node only aware of the addresses immediately before and after it in the chain.

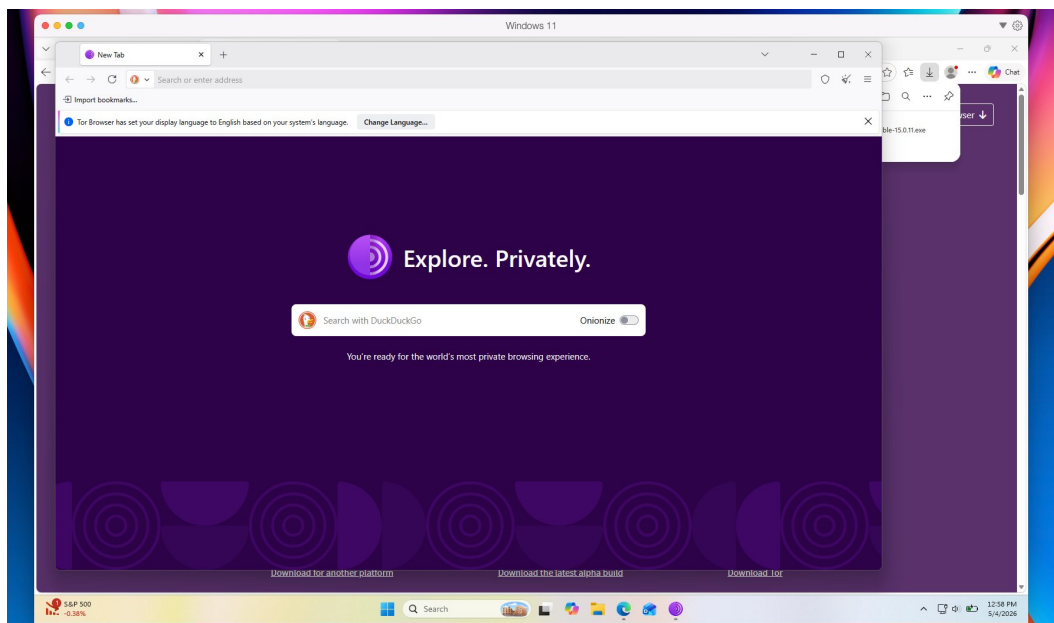


Figure 1: Tor Browser running and connected to the Tor network on the Windows 11 virtual machine.

Step 2: Privoxy Installation and Configuration

Privoxy version 4.1.0 was downloaded from SourceForge and installed on the virtual machine. After installation, Privoxy was confirmed to be running and listening on its default port of 8118. The main configuration file was edited using an Administrator Command Prompt with PowerShell to append the forwarding directive connecting Privoxy to the Tor network. The directive instructs Privoxy to forward all traffic using the SOCKS5t protocol to the Tor Browser local proxy on port 9150. The log directory was also configured to a path with appropriate write permissions.

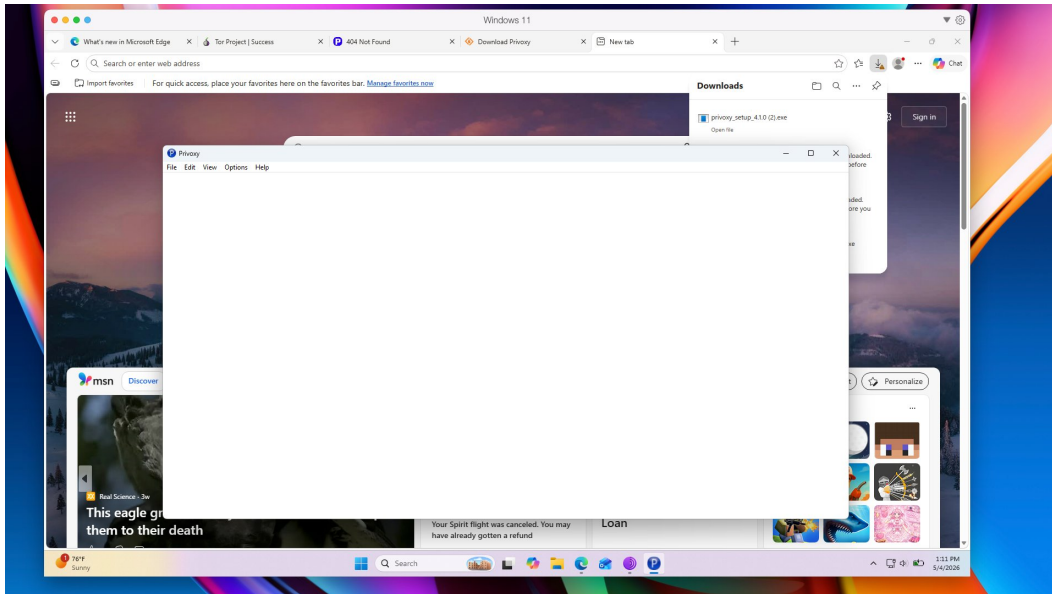


Figure 2: Privoxy successfully installed and running on the virtual machine.

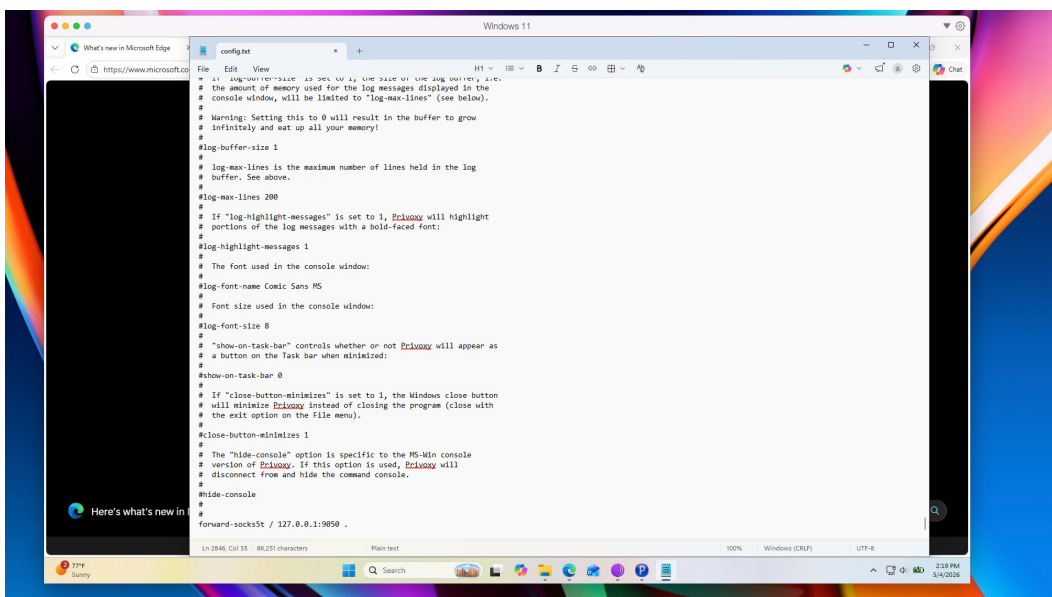


Figure 3: Privoxy configuration file showing the forward-socks5t directive added at the bottom.

Step 3: Firefox Proxy Configuration

Mozilla Firefox was installed and configured to route all traffic through Privoxy. In Firefox connection settings, manual proxy configuration was selected. The HTTP proxy was set to 127.0.0.1 on port 8118, and the SSL proxy was also configured to the same address and port to ensure encrypted HTTPS traffic was handled correctly. These settings were verified in Firefox advanced configuration, which confirmed that manual proxy mode was active and both the HTTP and SSL proxy values were correctly pointing to the Privoxy service.

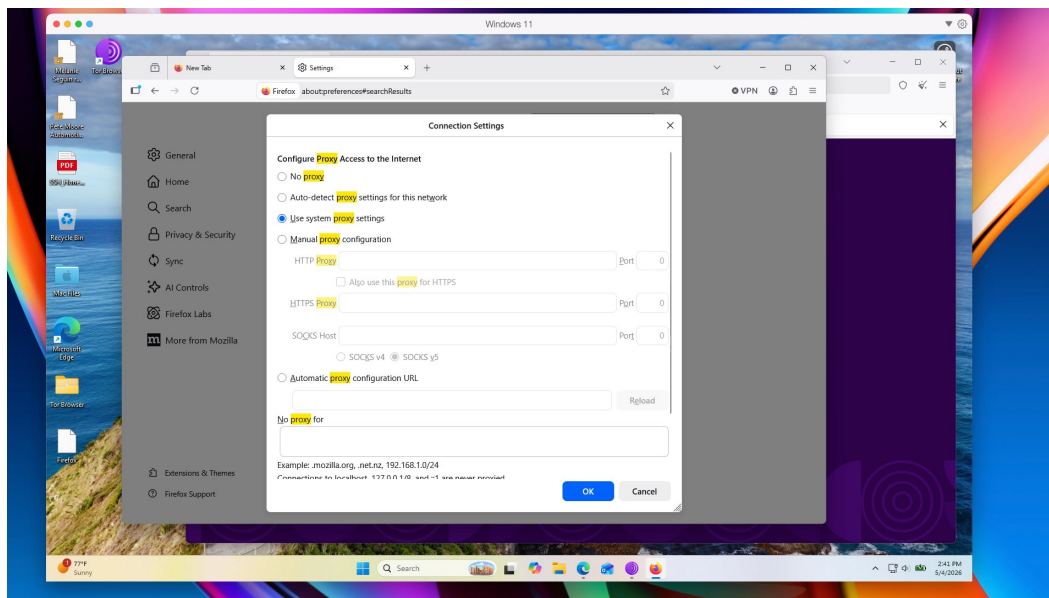


Figure 4: Firefox Connection Settings configured with manual proxy pointing to Privoxy on port 8118.

Step 4: Tor Routing Verification

The Tor routing chain was verified using two methods. First, the curl command line tool was used to send a request through Privoxy to the Tor Project IP check API, which returned a JSON response confirming the connection was identified as a Tor exit node with an IP address of 193.189.100.199. Second, Firefox was used to visit the Tor Project check page, which displayed the confirmation message: Congratulations. This browser is configured to use Tor, with the same Tor exit node IP address shown. This confirmed the complete chain from Firefox through Privoxy through Tor to the internet was functioning correctly.

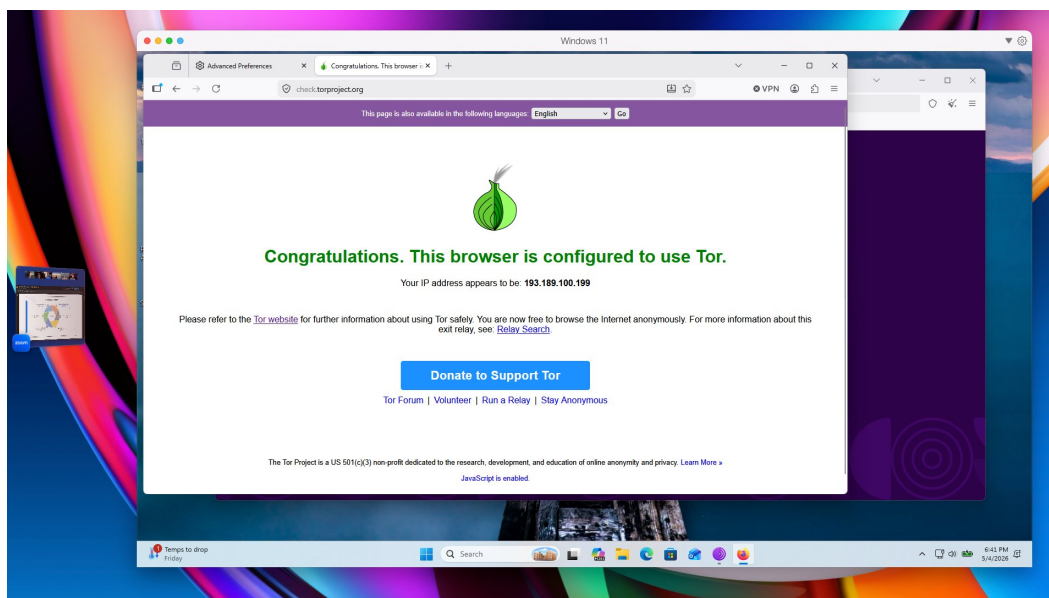


Figure 5: Tor Project confirmation page showing Firefox is successfully routing through the Tor network.

Step 5: Windows Firewall Configuration and Logging

An outbound firewall rule was created in Windows Defender Firewall with Advanced Security to block Firefox from making direct internet connections outside of the proxy chain. The rule was configured as a Program rule targeting the Firefox executable. The action was set to block the connection and the rule was applied across all three network profiles: Domain, Private, and Public. The rule was named Block Firefox Outbound and was confirmed active in the Outbound Rules list.

Firewall logging was also enabled across all three network profiles. Both dropped packets and successful connections were configured to be logged to the default firewall log path, ensuring a complete audit trail of all traffic processed by the firewall.

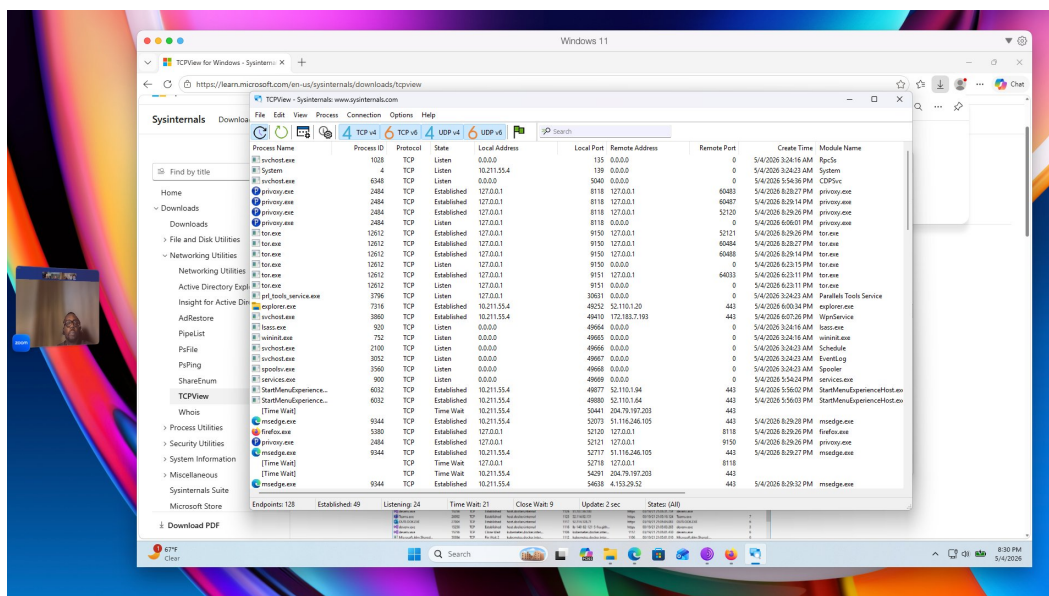


Figure 7: TCPView showing privoxy.exe, tor.exe, and firefox.exe connections confirming the full traffic chain.

Results and Conclusions

The lab was completed successfully. All web traffic originating from Firefox was anonymized through the Tor network via Privoxy, with the real IP address of the machine replaced by a Tor exit node IP address. Windows Defender Firewall was configured to block direct outbound connections from Firefox, enforcing the requirement that all traffic pass through the authorized proxy chain. Logging was enabled at both the firewall and proxy levels, and TCPView provided real-time confirmation that the traffic architecture was operating as designed.

This exercise builds directly applicable skills for Security Operations and Digital Forensics roles, particularly in the areas of network anonymization, proxy architecture, application-level firewall enforcement, and real-time traffic monitoring. The troubleshooting experience encountered throughout the lab, including Windows permission issues, configuration file path resolution, and proxy SSL settings, reflects the kind of problem solving required in real-world SOC environments.